



Examination Number:

Set:

Shore

Year 12

HSC Assessment Task 3

Term II Examination

2016

Mathematics Extension 1

General Instructions

- Reading time – 5 minutes
- Working time – 2 hours
- Write using black pen
- Board-approved calculators may be used
- The BOSTES Reference Sheet is provided separately
- Answer Questions 1–10 on the Multiple Choice Answer Sheet provided
- In Questions 11–14 show relevant mathematical reasoning and/or calculations
- Start each of Questions 11–14 in a new writing booklet
- Write your examination number on the front cover of each booklet

If you do not attempt a question, submit a blank booklet marked with your examination number and “N/A” on the front cover

Total marks – 70

Section I Pages 2–5

10 marks

- Attempt Questions 1–10
- Allow about 15 minutes for this section

Section II Pages 6–11

60 marks

- Attempt Questions 11–14
- Allow about 1 hour 45 minutes for this section

Section I

10 marks

Attempt Questions 1–10

Allow about 15 minutes for this section

Use the Multiple Choice Answer Sheet for Questions 1–10.

- 1 The point $P(x, y)$ divides the interval joining $A(-3, 4)$ to $B(7, 9)$ internally in the ratio $2:3$.

What are the coordinates of P ?

(A) $(1, 6)$

(B) $(1, 5)$

(C) $(0, 6)$

(D) $(0, 5)$

- 2 For what value of b is $(x+1)$ a factor of $x^3 - bx + 3$?

(A) $b = 3$

(B) $b = -3$

(C) $b = 2$

(D) $b = -2$

- 3 What is the acute angle between the lines $x + y = 1$ and $2x - y = 3$?

(A) 18°

(B) 19°

(C) 71°

(D) 72°

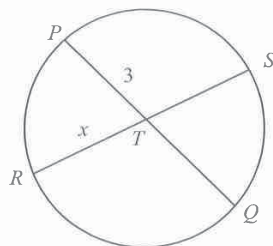
DO NOT REMOVE THIS PAPER FROM THE EXAMINATION ROOM

- 4 The point $P(x, y)$ moves so that it is equidistant from the y -axis and the point $S(2, 0)$.

What is the equation of the locus of P ?

- (A) $y^2 = 4(x - 2)$
 (B) $y^2 = 4(x - 1)$
 (C) $y^2 = 8(x - 2)$
 (D) $y^2 = 8(x - 1)$

5



NOT TO
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In the circle, chords PQ and RS intersect at T , $PQ = 7$, $PT = 3$, $RS = 8$ and $RT = x$.

What are the possible values of x ?

- (A) $x = 6$ or $x = 2$
 (B) $x = 4$ or $x = 2$
 (C) $x = 6$ or $x = 4$
 (D) $x = 4$ or $x = 3$
- 6 What is the domain for $f(x) = \sqrt{1-x} + \sqrt{1+x}$?
- (A) $x > 1$ or $x < -1$
 (B) $x \geq 1$ or $x \leq -1$
 (C) $-1 < x < 1$
 (D) $-1 \leq x \leq 1$

- 7 A curve has parametric equations $x = 3 \cos t$ and $y = 2 \sin t$.

What is the Cartesian equation of the curve?

- (A) $2x^2 + 3y^2 = 6$
 (B) $2x^2 - 3y^2 = 6$
 (C) $4x^2 + 9y^2 = 36$
 (D) $4x^2 - 9y^2 = 36$

- 8 The angle θ satisfies $\sin \theta = \frac{3}{5}$ and $\frac{\pi}{2} \leq \theta \leq \pi$.

What is the value of $\sin 2\theta$?

- (A) $\frac{24}{25}$
 (B) $-\frac{24}{25}$
 (C) $\frac{12}{25}$
 (D) $-\frac{12}{25}$

- 9 Which one of the following is the primitive (indefinite integral) of $\frac{1}{4x^2 + 16}$?

- (A) $\frac{1}{8} \tan^{-1} \frac{x}{2} + C$
 (B) $\frac{1}{4} \tan^{-1} \frac{x}{2} + C$
 (C) $\frac{1}{2} \tan^{-1} 2x + C$
 (D) $\frac{1}{4} \tan^{-1} 2x + C$

Section II

60 marks

Attempt Questions 11–14

Allow about 1 hour 45 minutes for this section

Start each of Questions 11–14 in a new writing booklet.

Question 11 (15 marks) Use a SEPARATE writing booklet

(a) Evaluate $\lim_{x \rightarrow 0} \frac{\sin x}{3x}$. 1

(b) Solve $\frac{2x}{1+x} \leq 1$. 2

(c) Find $\int \cos^2 x \, dx$. 2

(d) By using the substitution $\tan \frac{A}{2} = t$, or otherwise, show that 2

$$\frac{1 + \sin A + \cos A}{1 + \sin A - \cos A} = \cot \frac{A}{2}.$$

(e) Find $\frac{d}{dx} [\cos^{-1}(x^2)]$. 2

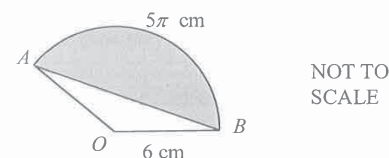
(f) (i) Write down the domain and range of the function $f(x) = 3 \sin^{-1}\left(\frac{x}{2}\right)$. 2

(ii) Sketch the graph of $y = f(x)$. 1

(g) Use the substitution $u = x^2 - 1$ to evaluate $\int_2^3 \frac{x}{(x^2 - 1)^2} \, dx$. 3

Question 12 (15 marks) Use a SEPARATE writing booklet

(a)



3

AOB is a sector of a circle with centre O .
The length of the arc AB is 5π centimetres.

Find the exact area of the segment cut off by the arc AB and the chord AB .

(b) Simplify $\frac{4 \sin 2\theta \cos 2\theta}{\cos 4\theta}$. 2

(c) (i) Express $\sin x + \sqrt{3} \cos x$ in the form $R \sin(x + \theta)$ 2
where $R > 0$ and $0 < \theta < \frac{\pi}{2}$.

(ii) Hence or otherwise solve the equation $\sin x + \sqrt{3} \cos x = \sqrt{3}$ for $0 \leq x \leq 2\pi$. 2

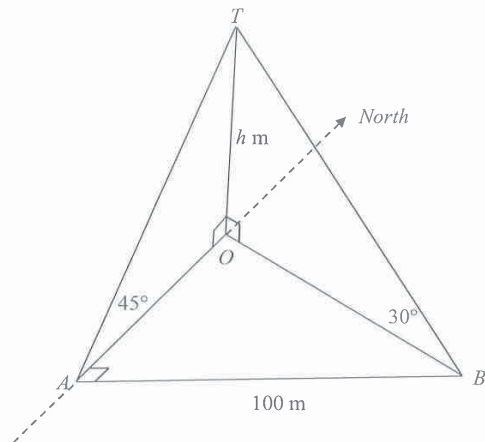
(d) Find the general solutions to $\cos\left(2x - \frac{\pi}{3}\right) = 0$. 3

(e) Prove by mathematical induction that for all integer values of $n \geq 1$, 3

$$\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \frac{1}{5 \times 7} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}.$$

Question 13 (15 marks) Use a SEPARATE writing booklet

(a)



A surveyor stands at a point A which is due south of a tower OT of height h metres. The angle of elevation of the top of the tower from A is 45° . The surveyor then walks due east for 100 metres to a point B , where he observes the top of the tower to have an angle of elevation of 30° .

- | | |
|--|---|
| (i) Show that $OB = h\sqrt{3}$. | 1 |
| (ii) Show that $h = 50\sqrt{2}$. | 2 |
| (iii) Find the bearing of B from the base of the tower.
Give your answer correct to the nearest degree. | 2 |

(b) The polynomial $P(x) = x^3 + px^2 + qx + r$ has roots $-\sqrt{\alpha}$, $\sqrt{\alpha}$ and β .

- | | |
|------------------------------------|---|
| (i) Show that $\beta + p = 0$. | 1 |
| (ii) Show that $\alpha\beta = r$. | 1 |
| (iii) Show that $pq = r$. | 2 |

Question 13 continues on page 9

Question 13 (continued)

(c) The points $P(8p, 4p^2)$ and $Q(8q, 4q^2)$ lie on the parabola $x^2 = 16y$.

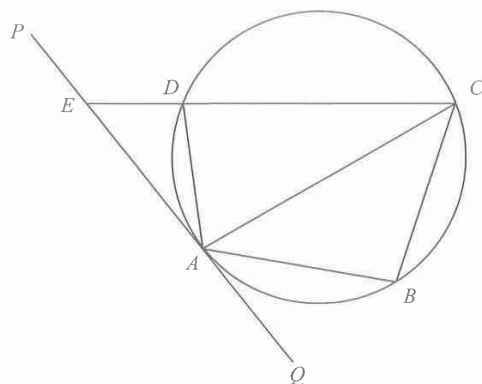
- | | |
|---|---|
| (i) Derive the equation of the chord PQ . Write your answer in general form. | 3 |
| (ii) Show that if PQ is a focal chord then $pq = -1$. | 1 |
| (iii) If PQ is a focal chord and P has coordinates $(4, 1)$, what are the coordinates of Q ? | 2 |

End of Question 13

Question 14 (15 marks) Use a SEPARATE writing booklet

- (a) In the diagram, PQ is a tangent to the circle at A .
Points B , C and D lie on the circle.

2



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Copy or trace the diagram into your writing booklet.

Show, giving reasons, that $\angle EAC = \angle ADE$.

- (b) (i) Find the range of the function $y = \frac{1}{\sqrt{4-x^2}}$,
(ii) Find the exact area between the curve $y = \frac{1}{\sqrt{4-x^2}}$, the x axis and the
lines $x = -\sqrt{3}$ and $x = \sqrt{2}$.

1

2

- (c) Let $f(x) = \sin^{-1}(-x) + \cos^{-1}(-x)$, where $-1 \leq x \leq 1$.

- (i) Show that $f'(x) = 0$.

2

- (ii) Hence deduce that $\sin^{-1}(-x) + \cos^{-1}(-x) = \frac{\pi}{2}$.

2

Question 14 continues on page 11

Question 14 (continued)

- (d) Given $f(x) = \log_e \left(\frac{2-x}{x} \right)$ for $0 < x < 2$.

- (i) Find $f'(x)$. Express your answer as a single fraction.

2

- (ii) Explain why $f(x)$ has an inverse function for $0 < x < 2$.

2

- (iii) Find the equation of the inverse function, $f^{-1}(x)$.

2

END OF PAPER

MID-YEAR EXAM SOLUTIONS

$$\begin{aligned} \textcircled{1} \quad P(x, y) &= \left(\frac{kx_2 + Lx_1}{k+L}, \frac{kx_1 + Ly_1}{k+L} \right) \\ &= \left(\frac{2(7) + 3(-3)}{2+3}, \frac{2(9) + 3(4)}{2+3} \right) \\ &= \left(\frac{5}{5}, \frac{30}{5} \right) \\ &= \underline{\underline{(1, 6)}} \quad \textcircled{A} \end{aligned}$$

(2) $P(-1) = 0 = (-1)^3 - 6(-1) + 3$
 $0 = -1 + 6 + 3$
 $b = -2$ (D)

$$\begin{aligned} \textcircled{3} \quad \tan \alpha &= \left| \frac{m_1 - m_2}{1 + m_1 m_2} \right| \\ &= \left| \frac{-1-2}{1+(-1)(2)} \right| \\ &= \left| \frac{-3}{-1} \right| \\ &= 3 \\ \alpha &= \tan^{-1} 3 \\ &= \underline{\underline{72^\circ}} \end{aligned} \quad \textcircled{D}$$

(4)  $y^2 = 4(x-1)$ (B)

⑤

$$PT \cdot TQ = RT \cdot RS$$
$$3 \times 4 = x(8-x)$$
$$12 = 8x - x^2$$
$$x^2 - 8x + 12 = 0$$
$$(x-6)(x-2) = 0$$
$$\underline{x = 6} \text{ or } \underline{x = 2}$$

Ⓐ

(6) $f(x) = \sqrt{1-x} + \sqrt{1+x}$

$1-x \geq 0$ and $1+x \geq 0$

$x \leq 1$ and $x \geq -1$

i.e. $\underline{-1 \leq x \leq 1}$ (D)

⑦ $x = 3 \cos t$ $y = 2 \sin t$

$$\frac{x}{3} = \cos t \qquad \frac{y}{2} = \sin t$$
$$\cos^2 t + \sin^2 t = 1$$
$$\left(\frac{x}{3}\right)^2 + \left(\frac{y}{2}\right)^2 = 1$$
$$\frac{x^2}{9} + \frac{y^2}{4} = 1$$

(C)

$$\underline{\underline{4x^2 + 9y^2 = 36}}$$

⑧ $\sin \theta = \frac{3}{5}$ $\cos \theta = -\frac{4}{5}$

$\sin 2\theta = 2 \sin \theta \cos \theta$

$= 2 \times \frac{3}{5} \times -\frac{4}{5}$

$= -\frac{24}{25}$ ⑧

$$\begin{aligned} \textcircled{9} \quad \int \frac{1}{4x^2+16} dx &= \frac{1}{4} \int \frac{1}{x^2+4} dx \\ &= \frac{1}{4} \times \frac{1}{2} \tan^{-1} \frac{x}{2} + C \\ \textcircled{A} &= \frac{1}{8} \tan^{-1} \frac{x}{2} + C \end{aligned}$$

(10) Odd function in 2nd quadrant
 $y = \frac{-x}{1+x^2}$ (D)

A ~~D~~ D B A D C B A D

Question 11

$$\begin{aligned} (a) \quad \lim_{x \rightarrow 0} \frac{\sin x}{3x} &= \lim_{x \rightarrow 0} \frac{\sin x}{x} \times \frac{1}{3} \\ &= 1 \times \frac{1}{3} \\ &= \underline{\underline{\frac{1}{3}}} \end{aligned} \quad [1]$$

$$(b) \quad \frac{2x}{1+x} \leq 1$$

Critical Points: $x \neq -1$, $\frac{2x}{1+x} = 1$

$2x = 1+x$
 $x = 1$

$\therefore -1 < x \leq 1$ [2]

$$\begin{aligned} \text{(c)} \quad \int \cos^2 x \, dx &= \int \frac{1}{2} (\cos 2x + 1) \, dx \\ &= \frac{1}{2} \left(\frac{1}{2} \sin 2x + x \right) + C \\ [2] \quad &= \frac{1}{4} \sin 2x + \frac{x}{2} + C \end{aligned}$$

$$\begin{aligned} (d) \quad LHS &= \frac{1 + \sin A + \cos A}{1 + \sin A - \cos A} \\ &= \frac{1 + \frac{2t}{1+t^2} + \frac{1-t^2}{1+t^2}}{1 + \frac{2t}{1+t^2} - \frac{1-t^2}{1+t^2}} \\ &= \frac{1+t^2+2t+1-t^2}{1+t^2+2t-(1-t^2)} \\ &= \frac{2t+2}{2t^2+2t} \\ &= \frac{2(t+1)}{2t(t+1)} \quad [2] \\ &= \frac{1}{t} \\ &= \cot \frac{A}{2} = \underline{RHS} \end{aligned}$$

$$\begin{aligned} (e) \quad & \frac{d}{dx} [\cos^{-1}(x^2)] \\ &= \frac{-1}{\sqrt{1-(x^2)^2}} \times 2x \\ &= \frac{-2x}{\sqrt{1-x^4}} \end{aligned} \quad [2]$$

$$\begin{aligned} (f) \quad f(x) &= 3 \sin^{-1}\left(\frac{x}{2}\right) \\ (i) \quad D: \quad &-1 \leq \frac{x}{2} \leq 1 \quad [2] \\ &\underline{\underline{-2 \leq x \leq 2}} \\ R: \quad &-\frac{\pi}{2} \leq \sin^{-1}\left(\frac{x}{2}\right) \leq \frac{\pi}{2} \\ &\underline{\underline{-\frac{3\pi}{2} \leq 3 \sin^{-1}\left(\frac{x}{2}\right) \leq \frac{3\pi}{2}}} \end{aligned}$$

(ii)

(1)

$$\begin{aligned} (g) \quad & \int_2^8 \frac{x}{(x^2-1)^2} dx \\ & u = x^2 - 1 \\ & du = 2x dx \\ & \frac{1}{2} du = x dx \\ & \underline{x=3 \rightarrow u=8} \\ & \underline{x=2 \rightarrow u=3} \\ & = \int_3^8 \frac{du}{2u^2} \\ & = \frac{1}{2} \left[\frac{u^{-1}}{-1} \right]_3^8 \\ & = -\frac{1}{2} \left[\frac{1}{u} \right]_3^8 \\ & = -\frac{1}{2} \left[\frac{1}{8} - \frac{1}{3} \right] \\ & = -\frac{1}{2} \times \frac{-5}{24} \\ & = \frac{5}{48} \end{aligned}$$

Question 12

(a) $l = r\theta$

$5\pi = 6\theta$

$\theta = \frac{5\pi}{6}$

$$\begin{aligned} \text{Area} &= \frac{1}{2}r^2(\theta - \sin\theta) \\ &= \frac{1}{2} \times 6^2 \left(\frac{5\pi}{6} - \sin\frac{5\pi}{6} \right) \\ &= 18 \left(\frac{5\pi}{6} - \frac{1}{2} \right) \\ &= \underline{\underline{(15\pi - 9) \text{ cm}^2}} \quad [3] \end{aligned}$$

(b) $\frac{4 \sin 2\theta \cos 2\theta}{\cos 4\theta}$

$$\begin{aligned} &= \frac{2 \times 2 \sin 2\theta \cos 2\theta}{\cos 4\theta} \\ &= \frac{2 \sin 4\theta}{\cos 4\theta} \quad [2] \\ &= \underline{\underline{2 \tan 4\theta}} \end{aligned}$$

(c)

(i) $\sin x + \sqrt{3} \cos x = R \sin(x + \theta)$

$$= R \sin x \cos \theta + R \cos x \sin \theta$$

$\therefore R \cos \theta = 1 \quad R \sin \theta = \sqrt{3}$

$$\cos \theta = \frac{1}{R} \quad \sin \theta = \frac{\sqrt{3}}{R}$$

$\sqrt{3} \quad R \quad \theta \quad 1$

$$\begin{aligned} R^2 &= 1^2 + (\sqrt{3})^2 \quad \cos \theta = \frac{1}{2} \\ &= 1 + 3 \\ &= 4 \\ \underline{R} &= \underline{2} \end{aligned}$$

$\therefore \underline{\underline{\sin x + \sqrt{3} \cos x = 2 \sin \left(x + \frac{\pi}{3} \right)}}$

[2]

(ii) $2 \sin \left(x + \frac{\pi}{3} \right) = \sqrt{3}$

$\sin \left(x + \frac{\pi}{3} \right) = \frac{\sqrt{3}}{2}$

$x + \frac{\pi}{3} = \frac{\pi}{3}, \frac{2\pi}{3}, \frac{7\pi}{3}, \dots$

for $0 \leq x \leq 2\pi \quad x = 0, \frac{\pi}{3}, 2\pi \quad [2]$

(d) $\cos \left(2x - \frac{\pi}{3} \right) = 0$

$2x - \frac{\pi}{3} = 2n\pi \pm \cos^{-1} 0$
(where n is an integer)

$2x - \frac{\pi}{3} = 2n\pi + \frac{\pi}{2} \quad 2x - \frac{\pi}{3} = 2n\pi - \frac{\pi}{2}$

$2x = 2n\pi + \frac{5\pi}{6} \quad 2x = 2n\pi - \frac{\pi}{6}$

$x = n\pi + \frac{5\pi}{12} \quad x = n\pi - \frac{\pi}{12}$

[3]

(e) $\frac{1}{1 \times 3} + \frac{1}{3 \times 5} + \dots + \frac{1}{(2n-1)(2n+1)} = \frac{n}{2n+1}$

Prove true for $n=1$

LHS = $\frac{1}{1 \times 3}$ RHS = $\frac{1}{2 \times 1 + 1}$

$= \frac{1}{3} \quad = \frac{1}{3}$

$\therefore$ true for $n=1$

Assume true for $n=k$

i.e. $S_k = \frac{k}{2k+1}$

Prove true for $n=k+1$

i.e. $S_k + \frac{1}{(2(k+1)-1)(2(k+1)+1)} = \frac{k+1}{2(k+1)+1}$

i.e. $S_k + \frac{1}{(2k+1)(2k+3)} = \frac{k+1}{2k+3}$

Q12 (e) continued

LHS = $\frac{k}{2k+1} + \frac{1}{(2k+1)(2k+3)}$

$= \frac{k(2k+3) + 1}{(2k+1)(2k+3)}$

$= \frac{2k^2 + 3k + 1}{(2k+1)(2k+3)}$

$= \frac{(2k+1)(k+1)}{(2k+1)(2k+3)}$

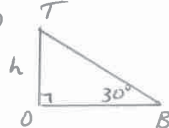
$= \frac{k+1}{2k+3}$

$= \text{RHS}$

$\therefore$ If true for $n=k$, then true for $n=k+1$.

By process of induction, true for all integers, $n \geq 1$. [3]

Question 13

(a) (i) 

$\frac{OB}{h} = \tan 60^\circ$

$OB = h \tan 60^\circ$

$= \underline{\underline{h\sqrt{3}}} \quad [1]$

(ii) In $\triangle OAT$, $OA = h$

In $\triangle OAB$, $OA^2 + 100^2 = OB^2$

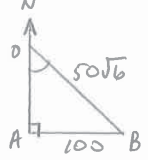
$h^2 + 10000 = (h\sqrt{3})^2$

$h^2 + 10000 = 3h^2$

$10000 = 2h^2$

$h^2 = 5000$

$h = \underline{\underline{50\sqrt{2}}} \quad [2]$

(iii) 

$OB = h\sqrt{3}$

$= 50\sqrt{2} \times \sqrt{3}$

$= \underline{\underline{50\sqrt{6}}}$

$\sin \angle AOB = \frac{100}{50\sqrt{6}}$

$= \frac{2}{\sqrt{6}}$

$\angle AOB = 55^\circ \quad [2]$

$\therefore \text{Bearing is } 180^\circ - 55^\circ = \underline{\underline{125^\circ}}$

(b) (i) $-\sqrt{x} + \sqrt{x} + \beta = \frac{-p}{1}$

$\beta = -p$

$\underline{\underline{\beta + p = 0}} \quad [1]$

(ii) $-\sqrt{x} \cdot \sqrt{x} \cdot \beta = \frac{-r}{1}$

$-\alpha \beta = -r$

$\underline{\underline{\alpha \beta = r}} \quad [1]$

(iii) $-\sqrt{x} \cdot \sqrt{x} + \sqrt{x} \cdot \beta + \sqrt{x} \cdot \beta = q$

$-\alpha = q$

$\underline{\underline{\alpha = -q}}$

sub into (ii) $-q \beta = r$

$\beta = \frac{r}{-q}$

sub into (i) $\frac{r}{-q} + p = 0$

$r - pq = 0$

$\underline{\underline{pq = r}} \quad [2]$

$$(c) (i) m = \frac{4p^2 - 4q^2}{8p - 8q}$$

$$= \frac{4(p-q)(p+q)}{8(p-q)}$$

$$= \frac{p+q}{2}$$

Eg'n of PQ is:

$$y - 4q^2 = \frac{p+q}{2}(x - 8q)$$

$$y - 4q^2 = \left(\frac{p+q}{2}\right)x - 4pq - 4q^2$$

$$\left(\frac{p+q}{2}\right)x - y - 4pq = 0 \quad [3]$$

(ii) PQ passes through (0, 4)

$$\therefore \left(\frac{p+q}{2}\right) \times 0 - 4 - 4pq = 0$$

$$-4pq = 4$$

$$\underline{pq = -1} \quad [1]$$

(iii) $P(8p, 4p^2) \equiv (4, 1)$

$$\therefore 8p = 4$$

$$p = \frac{1}{2} \rightarrow q = -2 \quad (pq = -1)$$

$\therefore Q$ is $(8x-2, 4(-2)^2)$

$$= \underline{(-16, 16)} \quad [2]$$

Question 14

$$(a) \angle EAC = \angle ABC$$

($\angle$ between chord + tangent equals $\angle$ in alternate segment)

$$\angle ABC = \angle ADE$$

($\angle$ in cyclic quadrilateral equals opposite exterior angle)

$$\therefore \underline{\angle EAC = \angle ADE} \quad (\text{both} = \angle ABC) \quad [2]$$

$$(b) (i) \text{ Range: } 4 - x^2 \leq 4$$

$$0 \leq \sqrt{4 - x^2} \leq 2$$

$$\underline{\underline{\frac{1}{\sqrt{4 - x^2}} \geq \frac{1}{2}}} \quad [1]$$

$$(ii) A = \int_{-\sqrt{3}}^{\sqrt{2}} \frac{1}{\sqrt{4 - x^2}} dx$$

$$= \left[\sin^{-1} \frac{x}{2} \right]_{-\sqrt{3}}^{\sqrt{2}}$$

$$= \sin^{-1}\left(\frac{\sqrt{2}}{2}\right) - \sin^{-1}\left(-\frac{\sqrt{3}}{2}\right)$$

$$= \frac{\pi}{4} - -\frac{\pi}{3}$$

$$= \underline{\underline{\frac{7\pi}{12} \text{ units}^2}} \quad [2]$$

$$(c) (i) f(x) = \sin^{-1}(-x) + \cos^{-1}(-x)$$

$$f'(x) = \frac{1}{\sqrt{1 - (-x)^2}} \times -1 + \frac{-1}{\sqrt{1 - (-x)^2}} \times -1$$

$$= \frac{-1}{\sqrt{1 - x^2}} + \frac{1}{\sqrt{1 - x^2}}$$

$$\underline{\underline{= 0}} \quad [2]$$

$$(c) (ii) \text{ If } f'(x) = 0$$

then $f(x) = \text{constant}$

$$\text{let } x=0 \quad f(0) = \sin^{-1}(0) + \cos^{-1}(0)$$

$$= 0 + \frac{\pi}{2}$$

$$= \underline{\underline{\frac{\pi}{2}}}$$

$$\therefore \underline{\underline{\sin^{-1}(-x) + \cos^{-1}(-x) = \frac{\pi}{2}}}$$

[2]

$$(d) (i) f(x) = \ln\left(\frac{2-x}{x}\right)$$

$$= \ln(2-x) - \ln x$$

$$f'(x) = \frac{-1}{2-x} - \frac{1}{x}$$

$$= \frac{-x - (2-x)}{(2-x)x}$$

$$= \underline{\underline{\frac{-2}{(2-x)x}}} \quad [2]$$

$$(ii) \text{ For } 0 < x < 2, (2-x)x > 0$$

$$\therefore \frac{-2}{(2-x)x} < 0$$

$$\text{i.e. } f'(x) < 0$$

i.e. $f(x)$ is monotonic decreasing

$\therefore f(x)$ has an inverse for $0 < x < 2$

[2]

$$(iii) \text{ let } y = \ln\left(\frac{2-x}{x}\right)$$

$$\text{Inverse} \Rightarrow x = \ln\left(\frac{2-y}{y}\right)$$

$$e^x = \frac{2-y}{y}$$

$$ye^x = 2-y$$

$$ye^x + y = 2$$

$$y(e^x + 1) = 2$$

$$y = \frac{2}{e^x + 1}$$

$$\therefore \underline{\underline{f^{-1}(x) = \frac{2}{e^x + 1}}}$$

[2]